

Problem

Which of these frequencies has the shorter period?

(a) 1 krad/s

(b) 1 kHz

Step-by-step solution

Step 1 of 2

(a)

Consider the frequency as,

$$\omega = 1 \text{ krad/s}$$

Determine the time period.

Consider the equation for time period.

$$\begin{aligned} T_{\text{rad/s}} &= \frac{2\pi}{\omega} \\ &= \frac{2\pi}{1000} \\ &= 6.283 \times 10^{-3} \\ T_{\text{rad/s}} &= 6.28 \text{ msec} \dots\dots (1) \end{aligned}$$

Step 2 of 2

(b)

Consider the frequency as,

$$\omega = 1 \text{ kHz}$$

Determine the time period.

Consider the equation for time period.

$$\begin{aligned} T_{\text{kHz}} &= \frac{1}{f} \\ &= \frac{1}{1000} \\ &= 1 \times 10^{-3} \\ &= 1 \text{ msec} \\ T_{\text{kHz}} &= 1 \text{ msec} \dots\dots (2) \end{aligned}$$

Compare equation 1 equation 2.

$$T_{\text{rad/s}} < T_{\text{kHz}}$$

Therefore, the 1 kHz frequency has the shorter time period.

Therefore, the correct option is **(b)**.